

SUPER CONDUCTIVITY

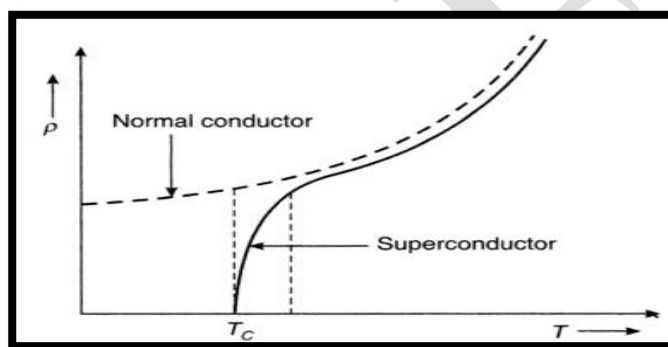
Introduction: Certain metals & alloys exhibit almost zero resistivity (infinite conductivity), when they are cooled sufficiently to low temperature is known as Super conductivity.

Super conductivity has got important applications like energy fields, energy technology, tele communications, computing, super fast transportation, medicine e.t.c

General properties:

The temperature at which the transition from normal stage to the Super conducting stage takes place on cooling in the absence of magnetic field is called critical temperature (T_C) is also known as Transition temperature.

H.K onnes by studying the electrical conductivity of metals he came across this phenomenon. He found that when pure mercury was cooled down below 4^0 K the resistivity suddenly dropped to zero. A similar behavior is observed in case of metal 'tin' the sudden drop in resistivity is graphically shown below.



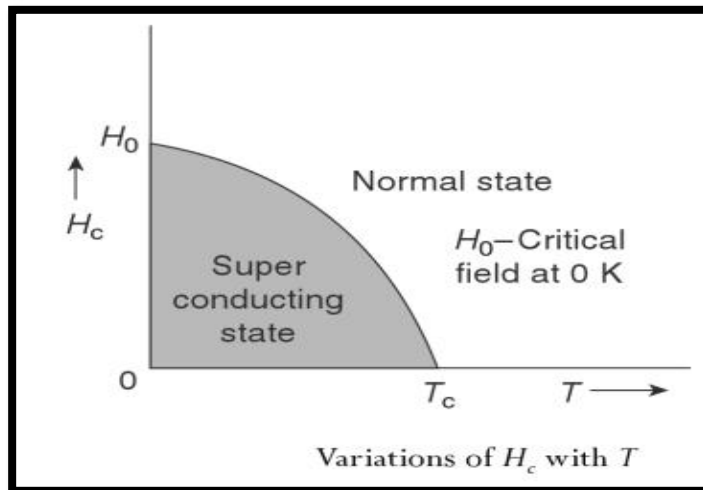
Properties observed:

1. T_C is different for different substances.
2. For chemically pure & structurally perfect Super conducting transition is very sharp.
3. Super conductivity substance in general lies in the inner column of periodic table.
4. Super conductivity is found in metallic substance in which the valence electron no lies in between 2&8.
5. From the periodic table it is observed that (T_C) of the elements & square of atomic no (Z^2) are having direct proportionality relation .
6. Transition metals having odd no of electrons exhibit Super conductivity.
7. Materials having high normal resistance exhibit Super conductivity.
8. Super conductivity is also observed in a substances having in all atomic volume with small atomic mass.
9. Ferro & anti Ferro magnetic substances do not exhibit Super conductivity.
10. In addition to dropping of electrical resistivity to zero the following changes are also observed.
 - a) The magnetic flux lines are rejected from the super conductor is known "MEISSNER EFFECT" .

- b) There is discontinuous change in specific heat.
- c) There are changes in thermal conductivity & volume of the substance.

Effect of magnetic field on super conductors:

Super conductivity disappears if the temperature of the specimen is raised above ' T_C '. In other words super conductivity is destroyed above ' T_C ', same effect is observed if the super conductor is subjected to magnetic field. In other words magnetic field destroys super conductivity. It can be indicated in graph below.



At temp $T = T_C$, $H_C = 0$. As T decreases, H starts increasing up to critical value (H_C), thus, applied magnetic field as the function of ' T ' can be written as

$$H_C(T) = H_C(0) \left[1 - \left(\frac{T}{T_C} \right)^2 \right] \text{-----} \rightarrow (1)$$

$H_C(0)$ ---- $\rightarrow$ Critical magnetic field at 0^0K

$H_C(0)$ & T_C ---- $\rightarrow$ are constants & also the characteristics of substance.

Numerical: A superconducting material has $T_C = 3.7\text{K}$ at zero magnetic field.

$H_C = 0.0306$ at 0K , find H_C at $T = 2\text{K}$ (0.02166T)

$$\text{Soln: } H_C(T) = H_C(0) \left[1 - \left(\frac{T}{T_C} \right)^2 \right]$$

$$H_C(T) = 0.0306 \left(1 - \left(\frac{2}{3.7} \right)^2 \right) = 0.0216591$$

Meissner effect:

When a weak magnetic field is applied to a super conducting specimen at a temperature below T_C , the magnetic flux lines are expelled from the specimen as shown in fig below:

The specimen acts as an ideal dia-magnet this is known as "meissner effect"

It is reversible, that when a temperature of specimen is raised below T_C at $T = T_C$, the flux lines suddenly start penetrating the specimen.

The specimen comes back to normal stage, under these conditions magnetic induction of a specimen $B = \mu_0 (H+I) \rightarrow (2)$

where, $H \rightarrow$ applied magnetic field.

$\mu \rightarrow$ magnetization.

As, $B=0$; $\mu_0 (H+I) = 0 \rightarrow (3)$

$$H = -I;$$

$$\chi = \frac{I}{H} = -1.$$

Hence susceptibility is -1, the substances are diamagnetic.

In case of super conductor resistivity ($\rho = 0$),

But electric field, $\varepsilon = J \rho \rightarrow (4) \Rightarrow \varepsilon = 0$.

Also from $\nabla \times E = \frac{dB}{dt} \rightarrow \frac{dB}{dt} = 0$; so, B should be constant.

This contradicts meissner effect itself.

Hence we can say that, super conducting substance or specimen behave differently from that of normal specimen. Ultimately, in super conducting stage $E=0$, & $B=0$

Effect of current:

To destroy the super conducting property critical magnetic field (H_C) is required. As the current is flowing through the super conductor the specimen can produce its own magnetic field if superconductivity is to be destroyed. In other words if the current in the specimen is increased to critical value I_C . So that H_C is produced, super conductivity disappears.

Hence, I_C can be related as $I_C = 2\pi r H_C \rightarrow (5)$

Mathematically I_C is related to H_C & is clear from following example.

Numerical:

Calculate critical current passing through the wires having diameter 1mm, at temp=4.2K, T_C for lead is 7.18K, value of $H_0 = 6.5 \times 10^4$ A/m.

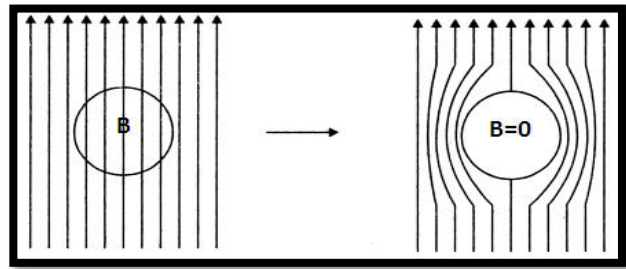
$$\text{Soln: } H_C = H_0 \left[1 - \left(\frac{T}{T_C} \right)^2 \right]$$

$$H_C = 6.5 \times 10^4 \left[1 - \left(\frac{4.2}{7.18} \right)^2 \right] = 42758.55246$$

$$I_C = 2\pi r H_C = 2 \times 3.14 \times 1 \times 42758.55246 = 134.518 \text{ A}$$

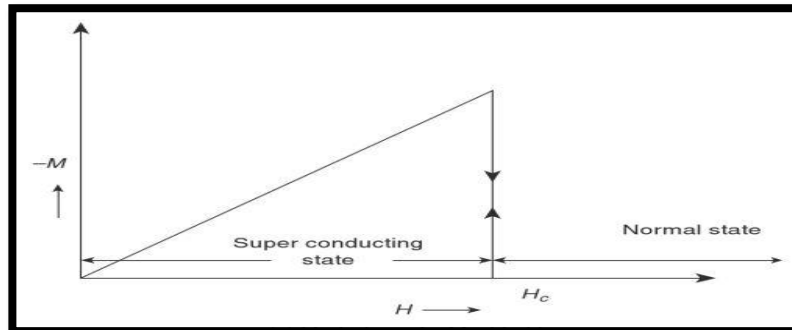
Types of super conductors:

They can be classified as Type-1 & Type-2 depending on the exhibition of meissener effect.

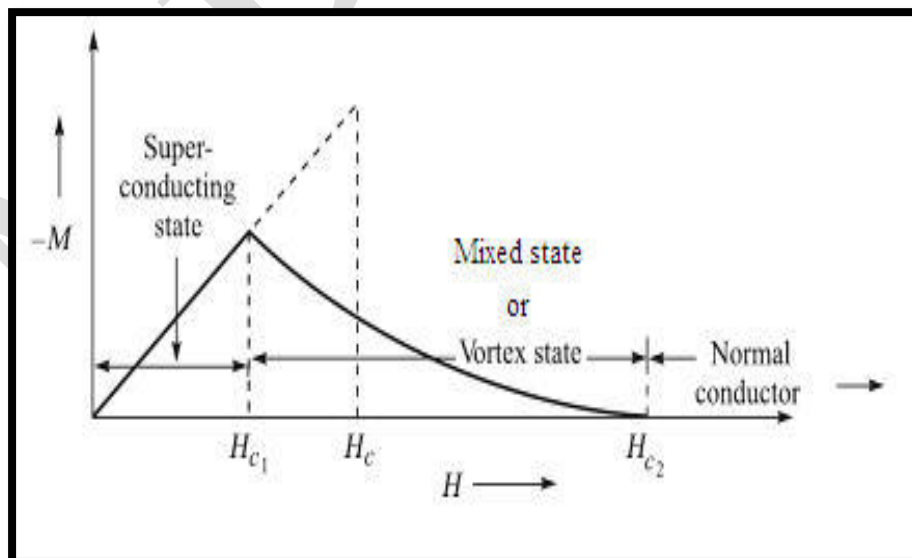


Type-1 Super conductor : They also known as soft super conductors ,when the magnetic field is applied & increased gradually from its initial value $H < H_c$, the diamagnetism disappears abruptly.

Super conducting specimen changes to normal state sharply as shown in fig. Al, zn, Hg are eg, of Type-1 Super conductor.



Type-2 Super conductor: As shown in fig, Type-2 conductors are having two critical fields namely H_{c1} & H_{c2} . Up to the field H_{c1} the specimen is in pure super conducting state rejecting all the magnetic lines. Beyond H_{c1} lower critical field ,magnetic lines starts penetrating the specimen ,up to the second critical field (H_{c2})(upper critical field),the specimen is in mixed state .In other words in this region between (H_{c1}) & (H_{c2}) the meissner effect is in complete .This region is “Vortex-region”. Beyond (H_{c2}) the specimen completely becomes normal from super conducting state. eg : Niobium (Nb) ,Zirconium (Zr) They also known as Hard superconductors, these have large practical applications, since they got very high current density.



UNIT V

NANO-TECHNOLOGY

Introduction:

Nanotechnology is defined as the study and use of structures between 1 nanometer and 100 nanometers in size. Nanotechnology is the engineering of functional systems at the molecular scale. This covers both current work and concepts that are more advanced. In its original sense, nanotechnology refers to the projected ability to construct items from the bottom up, using techniques and tools being developed today to make complete, high performance products.

Nanoscience: It deals with the study of properties of materials at nanoscales where properties differ significantly than those at larger scale. The applications of nanoscience emerged as nanotechnology.

Nano science deals with the synthesis, manipulation and the characterization of materials at atomic and molecular level and to study the various properties like electrical, magnetic, optical, mechanical and chemical etc.

Nanoscale:

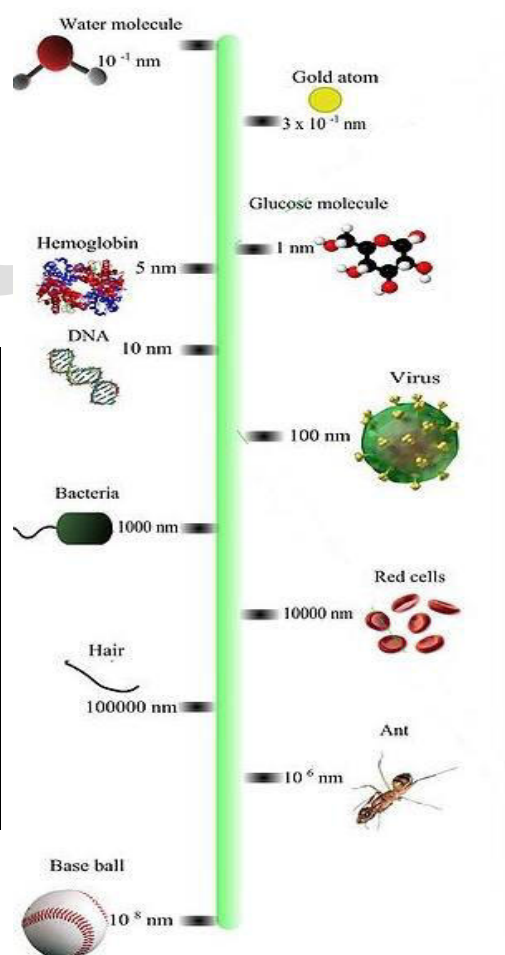
$$1\text{nm} = 10^{-9}\text{m} = 10^{-7}\text{cm}$$

Nano means 10^{-9}m i.e. a billionth part of a meter.

Atoms are extremely small & the diameter of a single atom can vary from 0.1 to 0.5nm depending on the type of the element.

Dimensions of few nanomaterials:

Carbon atom	0.15 in diameter
Water molecule	0.3nm
Red blood cell	7,000nm
Human hair	80,000nm wide
White blood cell	10,000nm
Virus	100nm
Hydrogen atom	0.1nm
Bacteria range	1,000 to 10,000nm
Proteins	5 to 50nm
DNA	2nm Width
Quantum dots	8nm



Nanotechnology: It deals with the design, characterization, production and application of nanostructures, nano-devices and nano-systems.

Nanomaterials:

The materials in which the atoms are arranged in the order of 1 to 100nm in any one of the dimension and these atoms will not move away from each other, called as nano materials. (All materials are composed of grains, which in turn comprise many atoms. The visibility of these grains depends on their size. The materials possessing grains of size ranging from 1 to 100nm, known as nanomaterial's, can be produced with different dimensionalities)

Ex: C, Zn, Cu – Fe alloys, Ni, Pd, Pt etc.

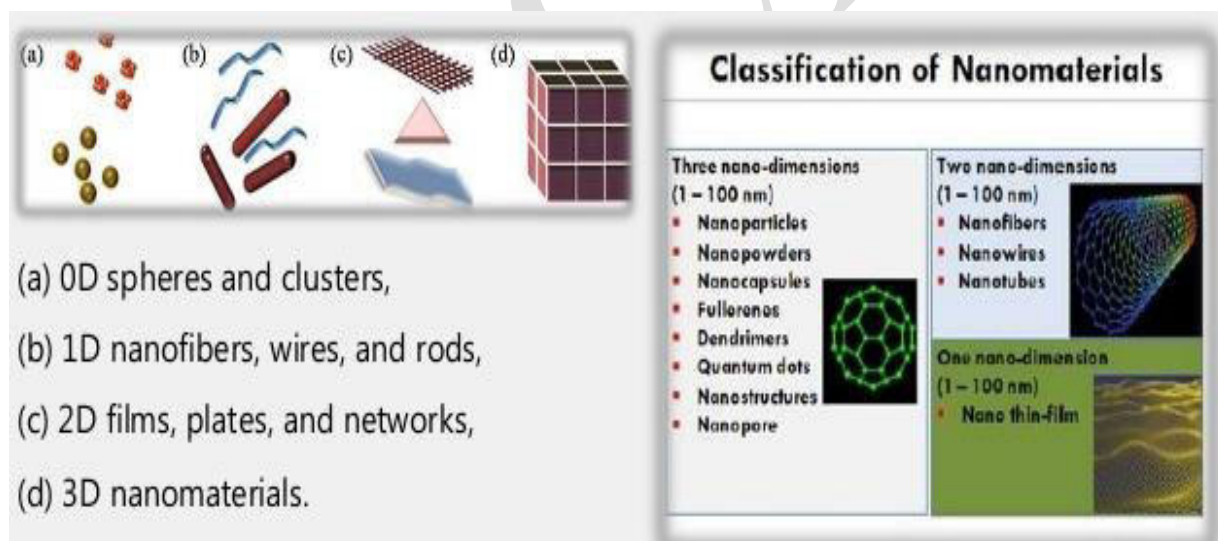
Zero-dimensional nanomaterials: Materials wherein all the dimensions are measured within the nanoscale (no dimensions, or 0-D, are larger than 100 nm). The most common representation of zero-dimensional nanomaterials are nanoparticles.

One dimensional nanomaterial: Materials that are nanoscale in one dimension called as one dimensional nano materials (nano layers). Examples: Nano tubes and nano wires.

Two dimensional nanomaterials: Materials having two of its dimension in nano scale is called two dimensional nano materials. Examples: Nano thin films, nano plates

Three dimensional nanomaterials: Materials having three of its dimension in nano scale is called three dimensional nano materials.

Examples: 3D particles of precipitates, Colloids, quantum dots, tiny particles of semiconductor materials.



Basic principles /properties of Nanomaterials:

When the material size of the object is reduced to nanoscale, then it exhibits different properties than the same material in bulk form. The factors that differentiates the nanomaterials from bulk material is

They are

- Increase in surface area to volume ratio
- Quantum confinement

Increase in surface area to volume ratio: Nano materials have a large surface area to volume ratio when compared to bulk material and it is a great measure for efficiency of nanotechnology.

Example: Let us consider a sphere of radius 'r'

$$\text{Its surface area} = 4\pi r^2$$

$$\text{Its volume (atom)} = \frac{4}{3}\pi r^3$$

$$\text{Surface area (S.A.) to volume ratio} = \frac{4\pi r^2}{\frac{4}{3}\pi r^3} = \frac{3}{r}$$

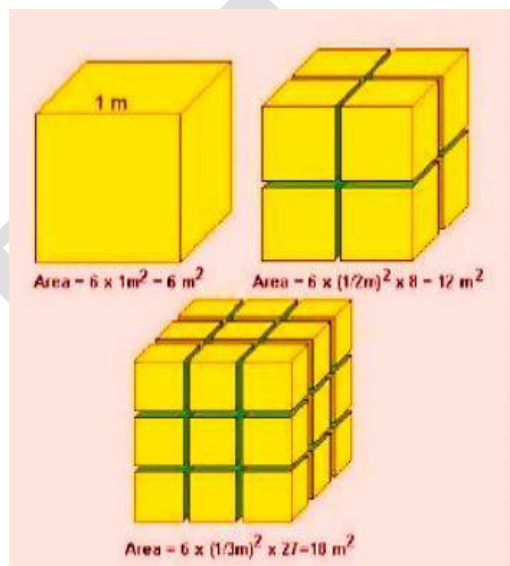
Therefore when the radius of the sphere decreases, its surface area to volume ratio increases.

Example: Let us consider a cubic volume shown in figure.

If a given sample its surface area is 6m^2 . When it is divided into 8 pieces its surface area becomes 12m^2 . When the volume is divided into 27 pieces its surface area becomes 18m^2 .

Thus, when the given volume is divided into smaller pieces, the surface area increases.

Due to increase in surface area, more number of atoms will appear at the surface. This makes nano materials more chemically reactive. Reducing the dimensions of material affects reactivity, optical properties, electrical properties, magnetic properties.



(ii) Quantum Confinement: The properties of materials can be studied based on the energy levels. When atoms are isolated, their energy levels are discrete. When very large no of atoms are closely packed to form a solid, the energy levels split & form bands. Nano materials represent intermediate stage. As a result, the energy levels change.

When we apply the problems of particles in a potential well as well as in a potential box. The dimensions of such wells or boxes are of the order of deBroglie wavelength of electrons, energy levels of electrons change. This effect is called Quantum confinement.

This affects the optical, electrical, magnetic properties of nanomaterials.

When the electrons are confined, the particles will have more oscillations and this will result in colour change of the materials. For examples, nano gold colloids are dispersed in ruby glass, the ruby glass exhibits red hue.

Synthesis (or) Fabrication of nanomaterials:

The production of nano materials or nano crystalline materials requires precise methods. There are various techniques that are capable of creating nano structures. In general there are two approaches that are classified as

- 1) Top-down approach (or) technique

2) Bottom-up approach (or) technique

Top-down approach: Top down techniques involves starting with a block of individual material, etching (removing the surface by dissolution) or convert it down to the desired size. The challenge here is to produce smaller and smaller structures. Nano material particle can be made through this method.

In this method, the nanomaterials are synthesized by assembling or arranging the bulk materials into nanosizes.

Top –down processing has been and will be the dominant process in semiconductor manufacturing.

Examples: (i) Sol-gel method.

(ii) Ball-milling method.

(iii) Lithography.

(iv) Mechanical grinding.

Bottom-up approach: Bottom-up technique involves the assembly of smaller sub unit (atoms or molecules) to make larger structure.

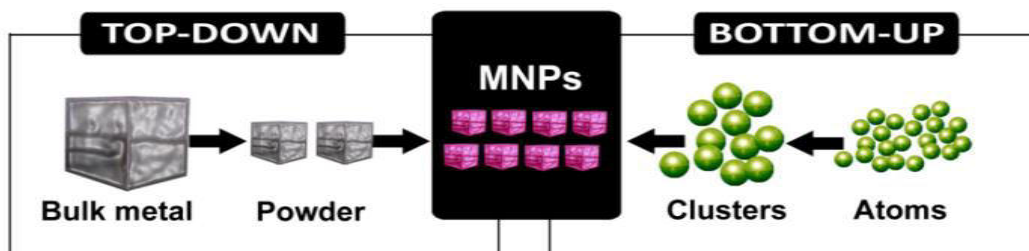
In this method, the nano materials are synthesized by assembling (or) arranging the atoms or molecules together to form the nano materials.

Examples: (i) physical vapour deposition method

(ii) Chemical vapour deposition method

(iii) Plasma arcing and

(iv) Electro deposition



Sol-Gel Method:

The sol-gel method is a wet chemical method or chemical solution deposition method. This technique is used to generate nanoparticles & nanopowders.

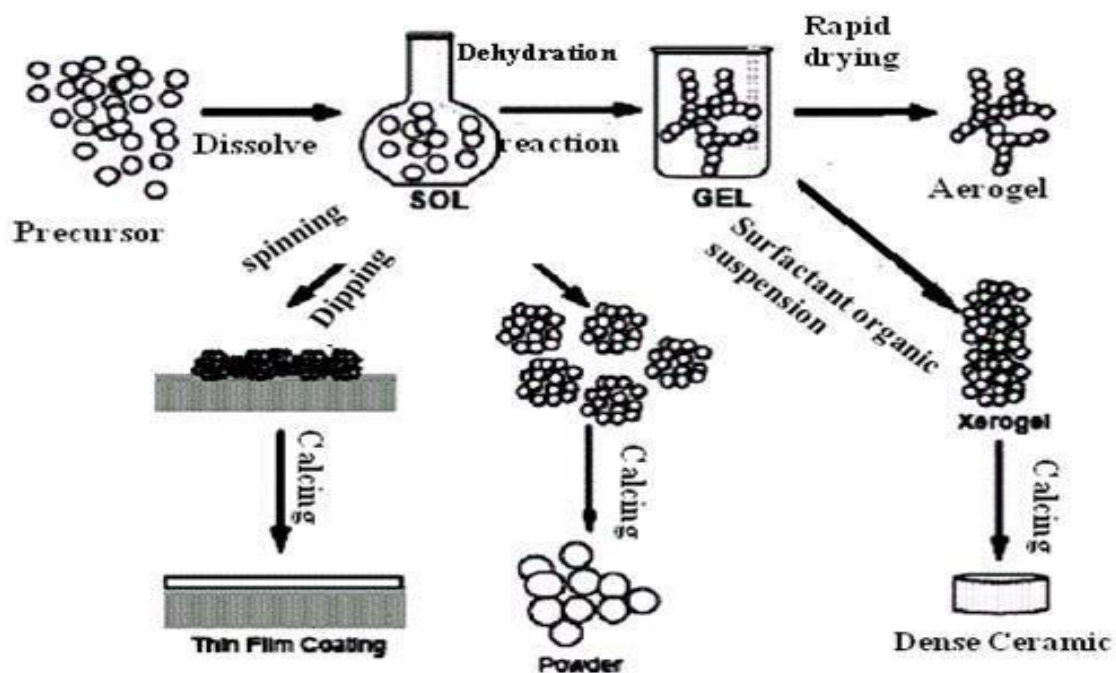
A given material converted into colloids & dissolved in water or in acids, then forms a solution (Sol). A colloid suspended in a liquid is called as “Sol”. A suspension that keeps its shape is called Gelatin or “Gel”. The sol-gel formation occurs in different stages. They are

1) Hydrolysis of precursors.

2) Condensation and polymerization of monomers to form particles.

3) Growth of particles and development of particles. (Agglomeration)

The schematic representation of the synthesis of nano particles using the Sol-Gel method is shown in fig.



This method takes place in following steps.

- (i) Take a material & convert it into liquid precursors (inorganic salts or organic species such as metal –alkoxide) & dissolved in water or in other solvents. It forms colloidal suspensions known as “Sol”.
- (ii) This solution is kept at a suitable temperature and some amounts of gelling agents are added to it. This will produce a gel. (By dehydration reaction with Sol forms Gel)
- (iii) Rapid drying of the gel, under super critical conditions an aero-gel.
- (iv) Drying of the “Gel” i.e. water & other liquids are removed from the gel forms a Xerogel. By calcination xerogel forms ceramics.
- (V) The solution further proceed through spinning & finally by calcination forms thin films & nanopowder respectively.

Advantages:

- (i) This method is used to prepare thin films, nanopowder, glasses, glass ceramics etc at very low temperatures.
- (ii) To prepare mono-sized nanoparticles.
- (iii) Very high purity in synthesized materials can be obtained.

Disadvantages:

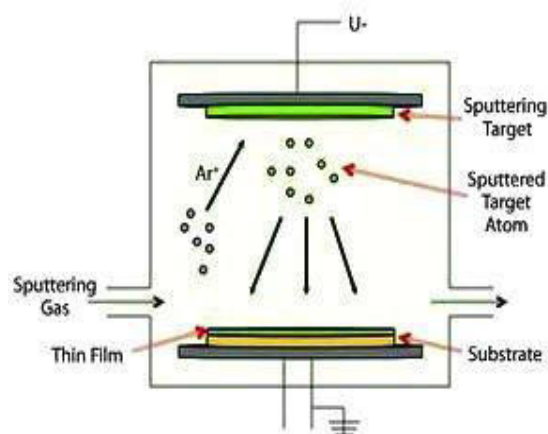
- (i) The raw materials are very costly.
- (ii) The synthesis reaction requires relatively longer time.
- (iii) Organic solvents used are harmful to the environment.

PVD(Physical vapour deposition technique):

Physical Vapour Deposition (PVD) is a collective set of processes used to deposit thin layers of material, typically in the range of few nanometers to several micrometers. PVD processes are environmentally friendly vacuum deposition techniques consisting of three fundamental steps :

- Vaporization of the material from a solid source assisted by high temperature vacuum or gaseous plasma.
- Transportation of the vapor in vacuum or partial vacuum to the substrate surface.
- Condensation onto the substrate to generate thin films.

Different PVD technologies utilize the same three fundamental steps but differ in the methods used to generate and deposit material.



The two most common PVD processes are **thermal evaporation** and **sputtering**.

Thermal evaporation is a deposition technique that relies on vaporization of source material by heating the material using appropriate methods in vacuum.

Sputtering is a plasma-assisted technique that creates a vapor from the source target through bombardment with accelerated gaseous ions (typically Argon).

In both evaporation and sputtering, the resulting vapor phase is subsequently deposited onto the desired substrate through a condensation mechanism to give nanofilms(thin-films).

Applications:

PVD is used in a variety of applications, & used in

- including fabrication of microelectronic devices,
- interconnects, battery and fuel cell electrodes,
- diffusion barriers,
- optical and conductive coatings,
- surface modifications.

Advantages:

- This method consists good strength and durability
- It is environment friendly vapor deposition technique.

Disadvantages:

- Cooling systems are required, to get nanomaterials.
- Mostly high temperature and vacuum control needs skill and experience.

Synthesis of Nanomaterials

Bottom- up Approach

1. Building up of nanomaterials from the bottom: atom-by-atom, molecule-by-molecule or cluster-by-cluster.
2. Method used are Sol-gel method, Hydrothermal synthesis.
3. There are less chances of contamination.

Top- down Approach

1. Process of miniaturizing or breaking down bulk materials into nanosized structures by mechanical deformation produced by ball milling.
2. Methods used are ball-milling, lithography.
3. The nanomaterials prepared by this process may be contaminated by tools and atmosphere.

6.3.1.BALL MILLING METHOD OF SYNTHESIS OF NANOMATERIALS:(Learn only this method)

Principle: Tiny rigid balls, usually, ceramic, flint pebbles and stainless steel, in a concealed container collide with each other to generate high pressure. The bulk material placed in this container is then crushed into nano crystal.

Construction and Working: Hardened steel or tungsten carbide balls are put in a container along with powder of particles ($50\mu\text{m}$) of a desired material as shown in figure 6.6. The container is closed with tight lids. When the container is rotating around the central axis, the material is forced to press against the walls. The milling balls impart energy on collision and produce smaller grains of nano-meter sizes. This process is used in producing metallic and ceramic nano materials. These mills are equipped with grinding media composed of wolfram carbide or steel.

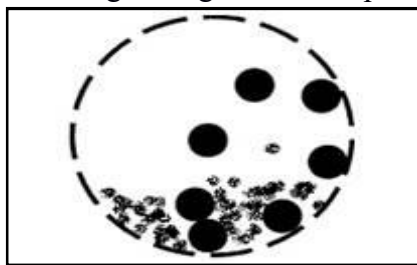


Fig.6.6: Ball milling Process

Advantages

- The significant advantage of this method is that it can be readily implemented commercially.
- Ball milling can be used to make carbon nanotubes and boron nitride nanotubes.
- It is a preferred method for preparing metal oxide nano crystals like Cerium(CeO_2) and Zinc Oxide (ZnO).

particles is 4.115 million times greater than the surface area of the silver dollar!

a) Electrical properties

The electrical properties of nanomaterials vary between metallic to semiconducting materials. It depends on the diameter of the nanomaterials. The very high electrical conductivity of nanomaterial is due to minimum defects in the structure.

b) Thermal conductivity

The thermal conductivity of nanomaterials are very high, is due to the vibration of covalent bonds. Its thermal conductivity is 10 times greater than the metal. The very high thermal conductivity of nanomaterial is also due to minimum defects in the structure.

c) Mechanical properties

Nanomaterials are very strong and withstand extreme strain. Most of the materials fracture on bending because of the presence of more defects, but nanomaterials possess only few defects in the structure.

Applications of Nanomaterials

Below we list some key applications of nanomaterials. Most current applications represent evolutionary developments of existing technologies: for example, the reduction in size of electronics devices.

a) Sunscreens and Cosmetics

The traditional chemical UV protection approach suffers from its poor long-term stability. A sunscreen based on mineral nanoparticles such as titanium dioxide offer several advantages. Titanium oxide nanoparticles have a comparable UV protection property. Nanosized titanium dioxide and zinc oxide are currently used in some sunscreens, as they absorb and reflect ultraviolet (UV) rays and yet are transparent to visible light and so are more appealing to the consumer. Nanosized iron oxide is present in some lipsticks as a pigment. The use of nanoparticles in cosmetics has raised a number of concerns about consumer safety.

b) Paints

Incorporating nanoparticles in paints could improve their performance, for example by making them lighter and giving them different properties. Thinner paint coatings ('**lightweighting**'), used for example on aircraft, would reduce their weight, which could be beneficial to the environment.

c) Displays

The huge market for large area, high brightness, flat-panel displays, as used in television screens and computer monitors, is driving the

development of some nanomaterials. Nanocrystalline zinc selenide, zinc sulphide, cadmium sulphide and lead telluride synthesized by sol gel techniques are candidates for the next generation of light-emitting phosphors.

d) Batteries

With the growth in portable electronic equipment (mobile phones, laptop computers, remote sensors), there is great demand for lightweight, high-energy density batteries. Nanocrystalline materials synthesized by sol-gel techniques are candidates for separator plates in batteries because of their foam-like (aerogel) structure, which can hold considerably more energy than conventional ones. Nickel-metal hydride batteries made of nanocrystalline nickel and metal hydrides are envisioned to require less frequent recharging and to last longer because of their large surface area.

e) Catalysis

In general, nanoparticles have a high surface area, and hence provide higher catalytic activity. Catalysis is important for the production of chemicals. Nanoparticles serve as an efficient catalyst for some chemical reaction, due to the extremely large surface to volume ratio. Platinum nanoparticles are now being considered in the next generation of automotive catalytic converters because the very high surface area of nanoparticles could reduce the amount of platinum required. Some chemical reactions are also carried out using nanomaterials. For example, reduction of nickel oxide to the base metal Ni.

f) Medicine

Nanotechnology has been a boon in medical field by delivering drugs to specific cells using nanoparticles. The overall drug consumption and side effects can be lowered significantly by depositing the active agent in the morbid region only and in no higher dose than needed. This highly selective approach reduces costs and human suffering. Nanotechnology can also help to reproduce or to repair damaged tissue. "**Tissue engineering**" might replace today's conventional treatments like organ transplants or artificial implants. For example, bones can be regrown on carbon nanotube scaffolds.

The use of gold in medicinal preparations is not new. In the Indian medical system called **Ayurveda**, gold is used in several preparations. One popular preparation is called **Saraswatharishtam**, prescribed for the memory enhancement. Gold is also added in certain medical preparations for babies in order to enhance their mental capability. Over 5000 years ago, Egyptians used gold in dentistry. In Alexandria, alchemists developed a powerful colloidal elixir known

as **liquid gold**, a preparation that was meant to restore youth. In china, people cook their rice with a gold coin in order to help replenish gold in their bodies.

g) Sensors of gases

The gases like NO₂ and NH₃ can be detected on the basis of increase in electrical conductivity of nanomaterials. This is attributed to increase in hole concentration in nanomaterials due to charge transfer from nanomaterials to NO₂ as the gas molecules bind the nanomaterials.

h) Food

Nanotechnology can be applied in the production, processing, safety and packaging of food. A nanocomposite coating process could improve food packaging by placing anti-microbial agents directly on the surface of the coated film. New foods are among the nanotechnology created consumer products coming onto the market at the rate of 3 to 4 per week. According to company information posted on PEN's Web site, the canola oil, by Shemen Industries of Israel, contains an additive called "**nanodrops**" designed to carry vitamins, minerals and phytochemicals through the digestive system and urea.

i) Construction

Nanotechnology has the potential to make construction faster, cheaper and safer. Automation of nanotechnology construction can allow for the creation of structures from advanced homes to massive skyscrapers much more quickly and at much lower cost. The Silica (SiO₂) is present in conventional concrete as part of the normal mix. When nano silica is added to concrete the particle packing can be improved mechanical properties. The addition of nano silica to cement based materials can also control the degradation of the fundamental C-S-H (**calcium silicate hydrate**) reaction of concrete caused by calcium leaching in water as well as block water penetration and therefore lead to improvements in durability. The strength of concrete can also be increase by adding haematite (Fe₂O₃) nanoparticles.

Steel has been widely available material and has a major role in the construction industry. The use of nanotechnology in steel helps to improve the properties of steel. The nano size steel produce stronger steel cables which can be used in bridge construction.

The glass is also an important material in construction. There is a lot of research being carried out on the application of nanotechnology to glass. Titanium dioxide (TiO₂) nanoparticles are used to coat glazing since it has sterilizing and anti-fouling properties. The particles catalyze powerful reactions which breakdown organic pollutants, volatile organic compounds and bacterial membranes. Most of glass in construction is

on the exterior surface of buildings. So the light and heat entering the building through glass has to be prevented. The nanotechnology can provide a better solution to block light and heat coming through windows.

Coatings is an important area in construction. Coatings are extensively use to paint the walls, doors and windows. Coatings should provides a protective layer which is bound to the base material to produce a surface of the desired protective or functional properties. Nanotechnology is being applied to paints to obtained the coatings having self healing capabilities and corrosion protection under insulation. Since these coatings are hydrophobic and repels water from the metal pipe and can also protect metal from salt water attack.

j) Agriculture

Applications of nanotechnology have the potential to change the entire agriculture sector and food industry from production to conservation, processing, packaging, transportation, and even waste treatment.

k) Energy

The most advanced nanotechnology projects related to energy are: storage, conversion, manufacturing improvements by reducing materials and process rates, energy saving and enhanced renewable energy sources. Today's best solar cells have layers of several different semiconductors stacked together to absorb light at different energies but they still only manage to use 40 percent of the Sun's energy. Commercially available solar cells have much lower efficiencies (15-20%). Nanotechnology could help increase the efficiency of light conversion by using nanostructures.

l) Other Applications

Some commercial products on the market today utilizing nanomaterials include stain resistant textiles and reinforced tennis rackets. Companies like Kraft foods are heavily funding nanomaterials based plastic packaging. Food will stay fresh longer if the packaging is less permeable to atmosphere. Coors Brewing company has developed new plastic beer bottles that stay cold for longer periods of time.

Conclusions

Nanotechnology is already transforming our lives with new technologies and products. Nanotechnology exploits the novel phenomenon and properties of matter at nano scale and may offer new solutions of millions of people in developing countries who lack access basic services, such as safe water, renewable energy, health care and education. There is a